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Zeta and L -Functions of Varieties and Motives

Bruno Kahn



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Introduction

Gauss described number theory as the queen of mathematics. Indeed, the amount of mathematics invented for arithmetic reasons is truly astonishing. To name just a few examples:

- A large part of complex analysis (Cauchy, Riemann, Weierstraß, Hadamard, Hardy–Littlewood, ...);
- The theory of divisors (Kummer, Dedekind) and of ideals (E. Noether);
- The theory of Riemann surfaces (Riemann, ...);
- The non-analytic version of the Riemann–Roch theorem for curves (F. K. Schmidt);
- The refoundation of Italian algebraic geometry on the basis of commutative algebra (Weil);¹
- The theory of abelian varieties (Weil);
- Part of the theory of linear representations of finite groups (Artin, Brauer);
- A large part of homological algebra: group cohomology, the theory of sheaves on a general site (Cartan, Eilenberg, Serre, Tate, Grothendieck, ...);
- Part of the theory of schemes (Grothendieck);
- The development of étale cohomology (M. Artin, Grothendieck, Verdier, Deligne...) and then crystalline cohomology (Grothendieck, Berthelot, Ogus, Bloch, Deligne, Illusie, ...);
- Part of the theory of derived categories (Grothendieck, Verdier);
- The six operations formalism (Grothendieck);
- Monodromy theory in the world of schemes (Grothendieck, Serre, Deligne, Katz, ...);

and, of course, the theory of motives!

¹ van der Waerden and Zariski also participated in this movement, for different reasons.

Zeta and L -functions are the meeting point of these theories: they crown the queen of mathematics. It is remarkable that a function whose definition is as simple as

$$\zeta(s) = \sum_{n=1}^{\infty} \frac{1}{n^s}$$

has played such a deep role, admitting vast generalisations which have shaped the evolution of number theory up to the present day; yet it remains the object of a conjecture whose analogue for varieties over finite fields was famously proved by Weil and Deligne, but whose original case due to Riemann remains an unapproachable mystery.

The text that follows is derived from a master's level course given at Jussieu in 2013. I have tried to give an exposition of part of the known results on zeta and L -functions, but also of the complex issues surrounding them, in an *ontogenetic* way: ontogenesis describes the progressive development of an organism from its conception to its mature form. With this goal in mind, I have scattered the text with quotations and commentaries which, I hope, will offer the reader a small window on the history of ideas in this field.

After having recalled in Chapter 1 the classical results (and hypothesis) on the Riemann zeta function, I introduce zeta functions of $\mathbf{Z}$ -schemes of finite type in the second chapter, which is essentially dedicated to the proof of the Riemann hypothesis for curves over a finite field. To my great regret, I was not able to do justice to Weil's proof of the Castelnuovo–Severi inequality in [Wei2]: to re-transcribe it in the language of schemes would have led me too far astray (see [64]). I only include in § 2.9.5 an idea of the proof, and treat the easy case of curves of genus 1 (due to Hasse). For the general case, I followed everyone else in giving the proofs of Mattuck–Tate and Grothendieck, which rely on an *a priori* weaker inequality; as a consolation prize, I compare the two inequalities in § 2.9.7 and show that we can recover the first one using the second and the additivity of Pic^τ (the numerically trivial divisors). A second proof of the Castelnuovo–Severi inequality via abelian varieties will be found in the next chapter (§ 3.2.2). It explains the first proof and puts it in a new light.

Chapter 3 is dedicated to the Weil conjectures. They are all proven, except for the hardest one: the “Riemann hypothesis”. I also give an overview of Dwork's p -adic proof of the rationality of zeta functions of varieties over a finite field (obtained before the development of Grothendieck's cohomological methods!).

Chapter 4 returns to more elementary mathematics, introducing Dirichlet, Hecke, and Artin L -functions. I give a proof of Dirichlet's theorem on arithmetic progressions, by the method expounded by Serre in [Ser3]; it would

however be a shame, as Pierre Charollois remarked to me, to omit Dirichlet's original method, which gave additional information and anticipated the analytic class number formulae (see Remark 4.22(2)). I then introduce the two generalisations of Dirichlet's L -functions: those of Hecke and Artin. I state without proof Hecke's main theorem: existence of an analytic continuation and a functional equation (Theorem 4.60), and then explain how Artin and Brauer derived the same results for non-abelian L -functions (Theorem 4.69).

The next chapter is the *pièce de résistance* of the text. It starts by introducing the approximate idea of Hasse–Weil L -functions, and ends by describing their precise definition due to Serre [113]. In the mean time, I explain the fundamental contributions of Grothendieck and Deligne: rationality and the functional equation for L -functions of l -adic sheaves in characteristic p , with essentially complete proofs; the theory of weights; and some theorems of Deligne on the Riemann hypothesis (the last of Weil's conjectures), this time without proof. Notably, one will find in § 5.4.4 an exposition of functional equations of the L -functions developed by Grothendieck in [CorrGS, letter from 30-9-64], and one will find in Theorem 5.58 a more precise statement and fairly complete parts of the proof of Grothendieck and Deligne's theorem on the rationality and the functional equation of Hasse–Weil L -functions in characteristic > 0 , confirming a conjecture of Serre (Conjecture 5.57) in this case.

The last chapter is dedicated to motives and their zeta functions. I limited myself to an elementary case: that of pure motives of Grothendieck, associated to smooth projective varieties over a finite field. One can go much further using the triangulated categories of motives introduced by Voevodsky and developed by Ivorra, Ayoub, and Cisinski–Déglise, but this would go beyond the scope of the book (see [63]). I explain how this viewpoint considerably clarifies how Weil cohomologies are used to prove rationality and the functional equation, and I do not resist the pleasure of applying this theory to prove a somewhat forgotten theorem of Weil: Artin's conjecture for non-abelian L -functions in positive characteristic (§ 6.15).

Finally, two appendices give supplements from categorical algebra. I also scattered the text with exercises, without attempting anything systematic.

Aside from Hecke theory, the automorphic side of the story is left essentially unaddressed; I contented myself with brief allusions here and there.

There still remains one question: what is the difference between a zeta function and an L -function? Morally, an L -function is a zeta function “with coefficients” (in a sheaf, in a representation, ...), and we recover a zeta function by taking trivial coefficients (cf. § 5.3). The situation becomes more complicated when we consider the Euler products associated by Hasse and Weil to a curve over a number field. . . . Another problem is which terminology

one should use in the case of motives. For pure motives, I have chosen that of zeta functions, which seems well established. It would probably be simpler to unify the terminology by dropping one of the two notations, but this is obviously delicate, for reasons of tradition.

I have benefited a great deal from previous expositions of the theory, from which I have borrowed much: these expositions would be too numerous to list here comprehensively, so I simply refer the reader to the bibliography. I thank the audience from the course, and in particular Matthieu Rambaud, for pointing out a number of misprints. Finally, I thank Joseph Ayoub, Pierre Charollois, Luc Illusie, Amnon Neeman, Ram Murty, and Jean-Pierre Serre for pertinent comments on this text.

Guide to the reader

This book is not a systematic exposition of a theory of zeta and L -functions – which does not exist, contrary to class field theory for example. I have chosen primarily to highlight the history of ideas. Thus, while certain proofs are complete, in others I prefer to emphasise the main ideas rather than give a complete but excessively technical proof; when such a proof is already available, I refer to expositions containing the missing parts.

Finally, there are cases where I have provided details that cannot be found in the literature. Here are some of them:

- § 5.2: a discussion of the notion of good reduction in relation to Hasse–Weil zeta functions (see in particular Proposition 5.4).
- § 5.4: Grothendieck’s functional equation, described in [CorrGS, letter from 30-9-64] (Theorems 5.29 and 5.30).
- Theorem 5.58, deducing from this functional equation the one described by Serre in [113, 4.1, example a)], with a precision that follows from “Weil II” (Deligne [30]).
- § 6.8: the theory of specialisation for pure motives.
- §§ 6.12 and 6.13: the zeta function of an endomorphism of a motive, its rationality, and its functional equation (see also § A.2.8 for an abstract categorical version, reproducing that of [60, I] in a slightly weaker form).
- § 6.15: a motivic proof of Artin’s conjecture in positive characteristic.
- § A.2.4: the notion of a “category with suspension”.

I have also recalled (without proof) several standard results, such as the Riemann–Roch theorem, the theory of abelian varieties, and the basics of algebraic number theory, for which there are already excellent expositions.

Finally, I assume some prerequisites, varying from one chapter to another:

- Chapters 1, 2 and §§ 4.1, 4.2: the basics of complex analysis.
- § 1.10 and Chapter 4: the basics of algebraic number theory, they can be found for example in the book of Lang [Lan2], in Cassels–Fröhlich [CF], among others.
- Chapters 2 and 3: the basics of algebraic geometry, as can be found for example in the book of Hartshorne [Har2]; the basics of intersection theory, as can be found in the book of Fulton [Ful].
- § 4.3: the basics of general topology.
- §§ 4.4 and 4.6: the basics of the theory of linear representations of finite groups, as in Serre’s book [Ser1].
- Chapter 5: much more serious foundations of the theory of schemes and étale cohomology, as developed in [SGA4, SGA4 $\frac{1}{2}$, SGA5] and [Mil]. This chapter is undoubtedly the hardest to read.
- Chapter 6: the basics of categorical algebra, as in the book of Mac Lane [McL].

Preface to the English edition

This version is a direct translation of the French version; it differs only by a few corrections and the addition of four exercises in Chapter 6. The numbering has also been changed.

1

The Riemann zeta function

The Riemann zeta function is defined to be

$$\zeta(s) = \sum_{n=1}^{\infty} \frac{1}{n^s}.$$

This function was first studied by Euler, and later by Riemann.

1.1 A bit of history

1.1.1 Euler

Euler established:

- $\zeta(2) = \frac{\pi^2}{6}$ (the Basel problem, 1735).
- The *Euler product*: $\zeta(s) = \prod_{p \text{ prime}} (1 - p^{-s})^{-1}$ (thesis, 1737).

He also obtained the more general formula

$$\zeta(2n) = (-1)^{n+1} \frac{B_{2n}(2\pi)^{2n}}{2(2n)!},$$

where the B_n denote the *Bernoulli–Seki numbers*¹.

1.1.2 Riemann ([95], 1859)

Riemann studied the function $\zeta(s)$ for $s \in \mathbf{C}$, establishing that:

¹ Seki Takakazu or Seki Kôwa was a Japanese mathematician and contemporary of Bernoulli, who discovered “Bernoulli numbers” in the same period as the latter, i.e. towards the end of the seventeenth century.

- It converges absolutely for $\Re(s) > 1$.
- It admits a meromorphic extension to $\mathbf{C}$, with a simple pole at $s = 1$ with residue $= 1$.
- It satisfies a functional equation (essentially conjectured by Euler in 1749): if we let $\Lambda(s) = \pi^{-s/2} \Gamma(s/2) \zeta(s)$, then

$$\Lambda(s) = \Lambda(1 - s).$$

- He also conjectured the Riemann hypothesis on the distribution of the zeros of $\zeta(s)$.

The functional equation gives a formula for the value of $\zeta(s)$ at negative integers (guessed by Euler):

$$\zeta(-n) = -\frac{B_{n+1}}{n+1} \quad (n > 0), \quad \zeta(0) = -\frac{1}{2}.$$

Riemann's motivation was the *prime number theorem* (on the distribution of prime numbers). This last result had been anticipated by Čebyšev (1848–1850) and was finally proved by Hadamard and de la Vallée Poussin (1896).

We will now prove some of these results. Part of the following exposition is borrowed from Serre [Ser3].

1.2 Absolute convergence

Proposition 1.1 *The function $\zeta(s)$ converges absolutely and uniformly on every compact set in the domain $\Re(s) > 1$, in which it defines a holomorphic function. It diverges at $s = 1$.*

Proof For both convergence and divergence, if $s = \sigma + it$ with $\sigma, t \in \mathbf{R}$, then we have $|n^{-s}| = n^{-\sigma}$, so we may suppose that $s \in \mathbf{R}$. We then use the following fact.

Lemma 1.2 (integral test) *Let $f : \mathbf{R}^+ \rightarrow \mathbf{R}^+$ be a decreasing function. Then $\sum_{n=1}^{\infty} f(n) < \infty \iff \int_1^{\infty} f(t) dt < \infty$.*

Proof We have the inequalities

$$f(n) = \int_n^{n+1} f(t) dt \geq \int_n^{n+1} f(t) dt \geq \int_n^{n+1} f(n+1) dt = f(n+1),$$

that is to say,

$$\int_{n-1}^n f(t) dt \geq f(n) \geq \int_n^{n+1} f(t) dt,$$

and we take the sum. □

This proof also gives the inequality

$$\left| \sum_{n=N}^{\infty} n^{-s} \right| \leq \int_{N-1}^{\infty} t^{-\Re(s)} dt = \frac{(N-1)^{1-\Re(s)}}{\Re(s) - 1},$$

whence uniform convergence; holomorphy follows from the next lemma, applied to the relevant partial sums. $\square$

Lemma 1.3 *Let U be an open subset of $\mathbf{C}$, and let $f_n : U \rightarrow \mathbf{C}$ be a sequence of holomorphic functions converging uniformly to f on every compact set in U . Then f is holomorphic on U , and the derivatives f'_n of f_n converge uniformly to the derivative f' of f on every compact set.*

Proof Let D be a closed disc in U with boundary ∂D , oriented counterclockwise, and let $z_0 \in \overset{\circ}{D}$. We apply the Cauchy formula to f_n :

$$f_n(z_0) = \frac{1}{2i\pi} \int_{\partial D} \frac{f_n(z)}{z - z_0} dz.$$

By passing to the limit, we obtain

$$f(z_0) = \frac{1}{2i\pi} \int_{\partial D} \frac{f(z)}{z - z_0} dz,$$

which shows that f is holomorphic on U , establishing the first part of the lemma. The formula

$$f'(z_0) = \frac{1}{2i\pi} \int_{\partial D} \frac{f(z)}{(z - z_0)^2} dz$$

gives the second part in the same manner. $\square$

1.3 The Euler product

Proposition 1.4 (Euler) *We have a factorisation*

$$\zeta(s) = \prod_p \frac{1}{1 - p^{-s}},$$

where the product, taken over all prime numbers, converges absolutely for $\Re(s) > 1$.

For the proof, say that a function $a : \mathbf{N} - \{0\} \rightarrow \mathbf{C}$ is *multiplicative* (resp. *completely multiplicative*) if it satisfies the identity $a(mn) = a(m)a(n)$ for all

coprime m, n (resp. for all m, n). The proposition follows from the next lemma applied to the function $a(n) = n^{-s}$.

Lemma 1.5 *Let a be a multiplicative function. Then the following conditions are equivalent:*

- (i) $\sum_{n=1}^{\infty} |a(n)| < +\infty$;
- (ii) $\prod_p (1 + |a(p)| + \cdots + |a(p^m)| + \cdots) < +\infty$.

If these conditions are satisfied, then

$$\sum_{n=1}^{\infty} a(n) = \prod_p (1 + a(p) + \cdots + a(p^m) + \cdots),$$

and if a is completely multiplicative:

$$\sum_{n=1}^{\infty} a(n) = \prod_p \frac{1}{1 - a(p)}.$$

Proof Suppose that (i) holds. In particular, $\sum a(p^m)$ converges absolutely for every p . For $x \in \mathbb{N}$, we then have

$$\sum_{n \in E(x)} a(n) = \prod_{p < x} \sum_m a(p^m),$$

where

$$E(x) = \{n \in \mathbb{N} - \{0\} \mid \text{all prime factors of } n \text{ are } < x\}.$$

Hence,

$$\left| \sum_{n=1}^{\infty} a(n) - \prod_{p < x} \sum_m a(p^m) \right| = \left| \sum_{n \notin E(x)} a(n) \right| \leq \sum_{n \geq x} |a(n)|$$

which shows that the infinite product converges to $\sum_{n=1}^{\infty} a(n)$. By applying this to $|a(n)|$, we see that the convergence is absolute, thus (ii) is satisfied.

Now suppose that (ii) holds. Then

$$\sum_{n < x} |a(n)| \leq \sum_{n \in E(x)} |a(n)| = \prod_{p < x} (1 + |a(p)| + \cdots + |a(p^m)| + \cdots),$$

thus (i) is satisfied. □

Corollary 1.6 *The function $\zeta(s)$ does not vanish for $\Re(s) > 1$.* □

Exercise 1.7 (Euler) Show that $\sum_{p \in \mathcal{P}} 1/p$ diverges.

1.4 Formal Dirichlet series

The notion of formal Dirichlet series is very useful for doing formal computations before thinking about questions of convergence.

Definition 1.8 (cf. [Eil, ch. 3, A4]) We define a *formal Dirichlet series* (with complex coefficients) to be an expression of the form

$$f = \sum_{n \geq 1} \frac{a_n}{n^s},$$

where n runs over the positive integers, and the a_n are complex numbers. If $f = \sum_{n \geq 1} \frac{a_n}{n^s}$ and $g = \sum_{n \geq 1} \frac{b_n}{n^s}$ are two formal Dirichlet series, we define their sum and their product as

$$f + g = \sum_{n \geq 1} \frac{a_n + b_n}{n^s},$$

$$fg = \sum_{n \geq 1} \frac{c_n}{n^s},$$

where $c_n = \sum_{pq=n} a_p b_q$.

Formal Dirichlet series form a unital commutative ring: the algebra of formal Dirichlet series with complex coefficients, denoted $\text{Dir}(\mathbf{C})$.

Remark 1.9 The algebra $\text{Dir}(\mathbf{C})$ is nothing other than the *total algebra*, sometimes also called the *large algebra* (following the terminology used by Bourbaki), of the multiplicative monoid $\mathbf{N} - \{0\}$ [BA3, ch. III, § 2, n° 10]. By writing each positive integer as a product of its prime factors, we see that $\text{Dir}(\mathbf{C})$ is isomorphic to the algebra of formal power series with complex coefficients on a countably infinite set of variables (cf. [BLie2, ex. 2 of the appendix]).

Definition 1.10 Let $f = \sum_{n \geq 1} \frac{a_n}{n^s}$ be a formal Dirichlet series. If $f \neq 0$, we define the *order* of f , denoted by $\omega(f)$, to be the smallest integer n such that $a_n \neq 0$. If $f = 0$, we set $\omega(f) = +\infty$.

Proposition 1.11 *The subsets $\{f \mid \omega(f) \geq N\}$ of $\text{Dir}(\mathbf{C})$ are ideals of $\text{Dir}(\mathbf{C})$. These subsets define a topology on $\text{Dir}(\mathbf{C})$ under which $\text{Dir}(\mathbf{C})$ is a complete topological ring.* $\square$

Corollary 1.12

- (1) *The sequence $(f_n)_{n \in \mathbf{N}}$ is summable in $\text{Dir}(\mathbf{C})$ if and only if $\lim_{n \rightarrow \infty} \omega(f_n) = +\infty$.*
- (2) *The sequence $(1 + f_n)_{n \in \mathbf{N}}$, where $\omega(f_n) > 1$ for all n , is multiplicable in $\text{Dir}(\mathbf{C})$ if and only if $\lim_{n \rightarrow \infty} \omega(f_n) = +\infty$.* $\square$

Lemma 1.5 gives:

Proposition 1.13 *Let $a : \mathbf{N} - \{0\} \rightarrow \mathbf{C}$ be a completely multiplicative function, and let $\alpha \in \mathbf{R}$. In the formal identity*

$$\sum_{n \geq 1} \frac{a(n)}{n^s} = \prod_p \left(1 - \frac{a(p)}{p^s}\right)^{-1},$$

the left-hand side converges absolutely for $\Re(s) > \alpha$ if and only if the right-hand side converges absolutely for $\Re(s) > \alpha$.

Exercise 1.14

- (a) Prove the following identity in $\text{Dir}(\mathbf{C})$:

$$\zeta(s)^{-1} = \sum_{n=1}^{\infty} \frac{\mu(n)}{n^s}, \quad \text{with}$$

$$\mu(n) = \begin{cases} (-1)^k & \text{if } n \text{ is the product of } k \text{ distinct prime numbers} \\ 0 & \text{otherwise.} \end{cases}$$

(The function $\mu(n)$ is called the *Möbius function*.)

- (b) Deduce from (a) the *Möbius inversion formula*: if $b_n = \sum_{d|n} a_d$, then $a_n = \sum_{d|n} \mu(d) b_{n/d}$.

More generally, let $f : [1, +\infty[\rightarrow \mathbf{C}$ be a function and $F(x) = \sum_{n \leq x} f\left(\frac{x}{n}\right)$. Then we have $f(x) = \sum_{n \leq x} \mu(n) F\left(\frac{x}{n}\right)$.

1.5 Extension to $\Re(s) > 0$; the pole and residue at $s = 1$

Proposition 1.15 *The function $\zeta(s)$ extends meromorphically to the domain $\Re(s) > 0$, with a simple pole at $s = 1$ with residue 1.*

To see this we need the following lemmas.

Lemma 1.16 *Let $(\alpha, \beta) \in \mathbf{R}^2$ be such that $0 < \alpha < \beta$. Let $z \in \mathbf{C}$ with $\Re(z) > 0$. Then we have*

$$|e^{-\alpha z} - e^{-\beta z}| \leq \left| \frac{z}{x} \right| (e^{-\alpha x} - e^{-\beta x}),$$

where $x = \Re(z)$.

Proof We write $e^{-\alpha z} - e^{-\beta z} = z \int_{\alpha}^{\beta} e^{-tz} dt$, and by passing to absolute values:

$$|e^{-\alpha z} - e^{-\beta z}| \leq |z| \int_{\alpha}^{\beta} e^{-tx} dt = \left| \frac{z}{x} \right| (e^{-\alpha x} - e^{-\beta x}). \quad \square$$

Lemma 1.17 (Abel's lemma) *Let (a_n) and (b_n) be sequences of complex numbers. Define $A_{m,p} = \sum_{n=m}^p a_n$ and $S_{m,m'} = \sum_{n=m}^{m'} a_n b_n$. Then we have*

$$S_{m,m'} = \sum_{n=m}^{m'-1} A_{m,n} (b_n - b_{n+1}) + A_{m,m'} b_{m'}. \quad \square$$

Proof of Proposition 1.15. Consider

$$\zeta_2(s) = \sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n^s}.$$

We apply Lemma 1.17 with $a_n = (-1)^{n+1}$, $b_n = n^{-s}$; Lemma 1.16 gives the inequality $|b_n - b_{n+1}| \leq \left| \frac{s}{\sigma} \right| (n^{-\sigma} - (n+1)^{-\sigma})$, with $\sigma = \Re(s) > 0$, therefore

$$\left| \sum_{n=m}^{m'-1} \frac{(-1)^{n+1}}{n^s} \right| \leq \left| \frac{s}{\sigma} \right| (m^{-\sigma} - m'^{-\sigma}) + m'^{-\sigma} \leq \left(\left| \frac{s}{\sigma} \right| + 1 \right) m'^{-\sigma}.$$

We conclude from this that ζ_2 converges for $\Re(s) > 0$. In addition,

$$\zeta(s) - \zeta_2(s) = 2^{1-s} \zeta(s) \Rightarrow \zeta_2(s) = (1 - 2^{1-s}) \zeta(s).$$

It follows that the function $(s-1)\zeta(s)$ extends meromorphically to $\Re(s) > 0$. At $s = 1$, we have $\zeta_2(s) = \log 2$. L'Hôpital's rule then gives

$$\lim_{s \rightarrow 1} (s-1)\zeta(s) = 1.$$